OBJCTIVES OF CURRENT PAPER

ZOO501- DEVELOPMENT BIOLOGY

Mid-Term File 2023 By

SULMAN ALI (BS-ZOOLOGY)

(0341-7547044)



1: Autonomous specification was first demonstrated in 1877 by a french medical students **laurent chabry**.chabry Desires to know causes of **birth** defects.

2:The division of embryo into **3 germ layers**.

3: **Teratology** is the study of birth defects.

4:The embryo contains **three germ layers** ectoderm,endoderm, mesoderm.

5:The individual of each species however, are divided into two mating types **minus and plus**

6: **fertilization** transmit genes from parents to offspring.

7:Play a major role in eye formation **Pax6.**

8:Allow the mRNA to exist the nucleus.**poly (A) tail.**

9: Enhancers are **DNA Sequences.**

10:The most crucial difference between mammalian cleavage all other types involve the phenomenon of **compation.**

11:The blastomere cell cycle is governed by the synthesis and degradation of **cyclia.**

12. cell type specification precede ……. Scale embryonic cell migration? (**large**)

13. Conditional mode of commitment involve6s interactions with…..? (**neighboring cells)**

14. Knowledge of gene activity in humans can be obtained by candidate…. (**gene mapping or Positional cloning.)**

15. sperm are attracted Toward eggs of their species by….. **(chemotaxis)**

16. …… contains little less Than half of the protein of the white, and most of the vitamins and minerals of the egg. (**yolk**)

17.Cortical ……. Are membrane-bound structures containing proteolytic enzymes. (**granules**)

18. finally, a fourth cortical granule protein ……. Forms a coating around the egg. (**hyalin**)

19. Upon fertilization, the intracellular ……. Concentration of the egg increases greatly. (**Ca ions)**

20. resting membrane Potential is generally about …… (**70 mV)**

21. Mouse fertilin is localized to the posterior plasma membrane of the sperm….. (**head**)

22. The zona pellucida is a ……which is synthesized and secreted by the growing oocyte. (**glycoprotein matrix)**

23. Fusion is an ……. Process, often mediated by specific "fusogenic" proteins. (**active**)

24. A.V Leeuwenhoek, the Dutch microscopist who co-discovered sperm in ……. (**1678**)

25. ……. Transcription factor, which is needed for mammalian eye, Nervous system, and pancreas development. (**Pax6**)

26. three germ layers are formed as a result gfof .. **gastrulation.**

27. During a process called cavitation, the trophoblast cells Secrete fluid into the morula to create…… **(blastocoel**)

28. The pattern of embryonic cleavage particular to a species is determined by …… major parameters. (**two**).

29:red blood cells make **globins**.

30:The zona pellucida is a **glycoprotein** matrix

31: Microfilaments formed by polymerization of **actin**

32: Resact diffuses in sea water and has a profound at very l**ow concentration**.

33: Envelope contain at least **eight** different glycoprotein

34: Coding regions of DNA IS CALled **exons**.

35: In **Anurans** frog and toad .

36; The **Regconition** interaction between sperm and egg.

37: upon fertilization, the intracellular …. Concentration ***calcium***

38: Activation of …. To start development ***metabolism***

39: Animal development is the division of embryo into three layers ectoderm, mesoderm.…***endoderm***

40: …**loss**. Of hydrogen ion causes pH to rise.

41: Germ layer are formed as the results of …. **Gastrulation**

42: Mesoderm generates are ….t**he kidney ,gonads ,bone , heart,and blood cells**

43: Spallenazed demonstrate that filtered …..**toad, semen devoid of sperm would not fertilize egg.**

44: coding regions of Dna …..**exons**

45: enhancers are….. **critical regulation of normal development.**

46: Between the blastoderm and yolk is species is called **germinal cavity**.

47: 3rd layers of development? **Endoderm**

48: Invariable cleavage leads **same** level

49:Tiny pores in egg shell?**17000**

50: The tail muscle cells from the **yellow** cresent

51: Of Hydrogen ions causes high pH? **Loss**

52: structural development of Avaian, reptilian and mammalian embryo is? **Primitive streak**

53: Cortical granules present below plasma membrane of egg? **15000**

54: Transcription factors? **3**

55: matrix of Zona pellucida made up of…….. matrix? **Glycoprotein**

56: Sperm discovered by A.V Leeuwenhoek in….? **1678**

57: functional motor unit of flagella? **Axoneme**

58: MPF promotes? **Mitosis**

59: The eggs of birds and fishes have only one small area of the egg that is free of yolk **(telolecithal eggs),**

60: Sea urchin contain …….. cleavage in zygote? **Radial**

61: .....Who first use the term ecology … **ernst Haeckel**..

62: ......when the ernst haeckel use the term .. **1869**

63:..... the grass land in Tropical region …**savana**..

64:…… in bones and teeth is **85%...** phosphorus.

65:....grass land on earth…**20**%..

66: steps of nitrogen fixation nitrification ,ammonification assimilation, all of these…answer **…all of these** .

67: One is effected the other is not .. **ammensalism**

68: sign for compition **..-/-…**

69: **Epilimnion** where the water surface face the solar heat and temperature increases.

70: **.. S and J shaped** both growth curves can be used.

71: coniferous forest characterised by conifers ..gymnospeem ..or both .. the ans is **both** ..

72: ..the land of mid night sun.. **tundra**..

73: hydro ..litho.. and atmosphere all include in ….. **biosphere**…

74: Chlamydomona usually haploid having…. Copies of chromosomes…(**Only one**).

75: pdx6 is specific for …..and endoderm.(**pancreatic**).

76:.....Protein synthesis is accomplished by ribosomes And….?(**tRNA**)

77:....Sea urchins exhibit which cleavage?..(**radial holoblastic**)

78:Mouse fertilin is localized to the posterior plasma Membrane of the sperm (**head**).

79: ...mesoderm forms…? (**Kidneys**)

80: ...MPF have how many units? (**2**)

81: The major cell adhesion ……Appear to be the Cadherins. (**Molecules**)

82 : The human β-globin gene has an enhancer in its 3′ UTR, roughly ……base pairs.(**700**)

83.The mammalian egg is also surrounded by a layer Of cells called….(**cumulus**)

84.Cell rearrange their ….. to form new immunoglobulins. **Dna**

85.Vitelline envelope consist of….. Glycoproteins ans : **8**

86:.Zp3 is specific glycoprotein in…. **mouse**.

87: Prechordial plate induce the formation of… **forebrain**

88:.Chlamydomona usually haploid having…. Copies of chromosomes…(**Only one).**

89: pdx6 is specific for …..and endoderm.(**pancreatic**).

90:.Protein synthesis is accomplished by ribosomes And….?(**tRNA**)

91: 4.Seaurchins exhibit which cleavage?..(**radial holoblastic**)

92:Mouse fertilin is localized to the posterior plasma Membrane of the sperm (**head**).

93:mesoderm forms…? (**Kidneys**)

94:MPF have how many units? **(2)**

95:The major cell adhesion …… Appear to be the Cadherins. (**Molecules**)

96:.The human β-globin gene has an enhancer in its 3′ UTR, roughly ……base pairs.(**700**)

97:The mammalian egg is also surrounded by a layer Of cells called….(**cumulus**)

98..Zp3 iska msqs tha,

99:..Syncytial…. **Insect**

100:Chlamydomonas. Which is usually Haploid, having just ***one*** copy of each chromosome

101….Eukaryotic genes are contained within a complex Of DNA and protein called ***chromatin***

102 ***Enhancers*** are critical in the regulation of normal Development.

103:Pancreatic And Duodenal Homeobox **1 gene**.

104:It is specific for the ***pancreatic*** region of the Endoderm.

105:***sea urchins*** contain 25,000 to 50,000 different types of mRNA

106..Mouse fertilin is localized to the **posterior plasma**

107:Membrane of the sperm ***head***

108: It appears that the acrosome-reacted sperm transfer Their binding from ZP3 to the adjacent ***ZP2** Molecules. .

109: Thus, fertilin appears to bind the sperm plasma Membrane to the ***egg plasma membrane***

110: This resting membrane potential is generally About ***70*** mV,

111…..Small subunit of MPF, the ****cyclin-dependent Kinase****.

112: **development biology** is a discipline that studies embryonic and other developmental processes.

113:**three** basic modes of commitment.

114:modes of commitment involves interactions with **neighboring cells.**

115:Nuclear control of a cell morphogenesis and the interaction of nucleus and cytoplasm Are well studied in **Acetabolaria**.

116: Metamorphosis is initiated by hormones from the tadpole **thyroid gland**.

117:The sperm to the egg by binding **6beta1 integrin protein** on the egg plasma membrane.

118: The mamalian egg is also surrouned by a layer of cells called the **cumulus.**

119: the second major group of protostome is **Lophotrochozoa**.

120: paired box gene is **6.**

121:A.V leeuvenhoek, the Dutch microscope

Who codiscovers sperms in **1678.**

122: These eggs have sparse equally space yolked and are **isolecithal equal yolk.**

123: The sperm travel by the amoeboid motion of lamellipodial extensions of the__ **cell membrane**

124: Mouse fertilin is localized to the posterior plasma membrane of the sperm **head**

125: Modes of commitment **3**

126: __ Allow the mRNA to exist the nucleus. **poly (A) tail**

127: cortical granule protein ……. Forms a coating around the egg. **(hyalin)**

128: ___Transcription factor, which is needed for mammalian eye, nervous system, and pancreas development. **(Pax6)**

129: Zp3 is specific glycoprotein in…. **mouse**

130: Chlamydomona usually haploid having…. Copies of chromosomes **one .**

131: Tunicate gastrulation is characterized by the invagination of the endoderm, the involution of the **mesoderm** , and the epiboly of the ectoderm.

132: Insect such as drosophila use **All Three Modes Of Specification**

133: All multicellular organisms arise by a slow process of progress change called .... **Development**

133: Coding regions of DNA:------ **Exons**

134: Annelids, molluscs and flatworms belong to:------- **Lophotrochozoan**

135: The blastula stage of sea urchin has __ **128-cell**

136: 14 amino acid peptide called **resact** ,

137: The __ is achieved by changing the electric potential of the egg plasma membrane **Fast block polyspermy**

138: Small subunit of MPF is the cyclin-dependent **kinase**

139: a region of globular __ molecules lies between the nucleus and the acrosomal vesicle. **Actin**

140: Metamorphosis is initiated by hormones from the tadpole **thyroid gland**.